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Superposed Bateman Biexponentials with Clearance-Constrained Scaling and Per-User Calibration:

A Pharmacokinetic Framework for Hormone Replacement Therapy Monitoring in the Isocline Mobile Application

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Abstract

We present the pharmacokinetic (PK) model underlying Isocline, a mobile application that predicts serum hormone concentration over time from a user's logged dose history for gender-affirming hormone replacement therapy (HRT). The model employs a one-compartment Bateman biexponential—first-order absorption from an intramuscular depot into a single well-mixed compartment, first-order elimination from it—with linear superposition of all prior dose contributions, a dynamic endogenous baseline that decays under gonadal suppression, and a per-user calibration factor α derived from self-reported blood tests. Ten medication/route/form combinations are supported: estradiol (valerate, cypionate, oral, transdermal) and testosterone (cypionate, enanthate, propionate, undecanoate, oral, transdermal).

The scaling constant S—an empirical surrogate for F/V_d—is derived in closed form for testosterone cypionate from a primary population-PK source (Bi et al. 2018), and for the remaining injectable testosterone esters by a conservation-of-mass identity that constrains S to depend only on the elimination rate k_e and testosterone clearance, independent of the ester's half-life. Estradiol ester constants are anchored to compiled steady-state PK tables. Transdermal routes are modeled as 24 hourly micro-doses approximating square-wave release.

We report accuracy against published anchors (0.0–0.1% for anchored quantities), out-of-sample validation (1–15% for estradiol and testosterone single-dose and steady-state checks), and time-to-peak ($t_{\max}$) agreement with primary and secondary sources. We quantify structural limitations: testosterone undecanoate's peak/trough ratio (8.1$\times$ modeled vs. 1.8$\times$ observed, a source conflict rather than a structural deficiency), transdermal intraday ripple ($C_{\max}/C_{\min} \approx 1.05$ vs. 2.24 observed), and oral testosterone peak fluctuation under-prediction.

This revision additionally reports three defects found by re-deriving the model from its sources rather than from its own prior statements. The endogenous baseline is keyed to the hormone being tracked rather than to the user's gonads, over-stating a transmasculine user's endogenous testosterone by more than an order of magnitude at treatment onset. The per-user calibration factor is derived against total predicted level but applied to the exogenous term alone, so it fails to reproduce the measurement that produced it—by up to 20% in exactly the regime where the baseline is largest, and where the two defects compound. And the spironolactone clearance adjustment has no supporting human pharmacokinetic study and is applied to the absorption exponential rather than the disposition one. All three are stated with their magnitudes rather than repaired in the figures. The model is honest about where its constants are primary-sourced, where they are secondary, and where they remain uncalibrated.

Keywords: pharmacokinetics, Bateman function, hormone replacement therapy, testosterone cypionate, estradiol valerate, flip-flop kinetics, superposition, per-user calibration, mobile health

Contents

1 Introduction **4**
1.1 Contributions . . . 4

2 Methods **4**
2.1 System overview . . . 4
2.2 Governing equation . . . 5
2.3 Time to peak . . . 6
2.4 Flip-flop kinetics . . . 6
2.5 Effective dose factor . . . 6
2.5.1 Ester weight factor . . . 6
2.5.2 Bioavailability . . . 7
2.6 The scaling constant . . . 7
2.7 Testosterone cypionate: primary-source derivation . . . 7
2.7.1 Rate constants . . . 7
2.7.2 Why 3.84 days and not 7–8 . . . 8
2.7.3 Scaling constant in closed form . . . 8
2.7.4 Time to peak . . . 8
2.8 Clearance identity for TE, TP, and TU . . . 8
2.8.1 Derivation . . . 8
2.8.2 The load-bearing check . . . 9
2.8.3 Application to TE, TP, and TU . . . 9
2.8.4 Steady-state predictions at typical regimens . . . 9
2.9 Testosterone enanthate and undecanoate: absorption rates . . . 9
2.9.1 Testosterone enanthate (TE) . . . 9
2.9.2 Testosterone undecanoate (TU) . . . 10
2.9.3 Testosterone propionate (TP) . . . 10
2.10 Estradiol ester derivations . . . 11
2.10.1 Estradiol valerate (EV) and cypionate (EC) . . . 11
2.10.2 Oral estradiol . . . 11
2.10.3 Transdermal estradiol . . . 11
2.11 Transdermal testosterone . . . 11
2.12 Oral testosterone undecanoate . . . 11
2.13 The basal offset used in the testosterone comparisons . . . 12
2.14 Dynamic endogenous baseline . . . 12
2.15 Blood test calibration . . . 13
2.15.1 The calibration factor . . . 13
2.15.2 Application, and a round-trip defect . . . 13
2.15.3 Hormone-type gating . . . 13
2.16 Drug interactions: spironolactone . . . 14
2.17 Aromatization estimation . . . 14
2.18 Chart window and sampling resolution . . . 15

3 Encoded constants **15**

4 Results **16**
4.1 Accuracy against anchors . . . 16
4.2 Out-of-sample validation . . . 16
4.3 Time to peak . . . 17
4.4 Persona validation . . . 17

5 Discussion **18**
5.1 Structural limitations 18
5.1.1 Testosterone undecanoate peak/trough ratio 18
5.1.2 Transdermal intraday ripple 18
5.1.3 Transdermal dose-response sublinearity 19
5.1.4 Oral testosterone peak fluctuation 19
5.1.5 Testosterone propionate remains uncalibrated 19
5.1.6 The endogenous baseline is keyed to the wrong variable 19
5.1.7 Model-wide simplifications 20
5.2 The steady-state vs. single-dose anchor choice 20
5.3 Source quality hierarchy 21
5.4 The per-user calibration factor as a pragmatic IIV analog 21

6 Conclusion **22**

1 Introduction

Gender-affirming hormone replacement therapy (HRT) involves the administration of exogenous hormones—estradiol and/or testosterone—to align an individual's serum hormone levels with their gender identity. Clinical management relies on periodic blood draws to measure serum hormone concentration, titrate dosage, and confirm therapeutic levels [11]. Between draws, patients and providers have limited tools to anticipate where levels are heading, when a peak or trough is expected, or how a missed dose or regimen change will affect the trajectory.

Isocline is a mobile application that addresses this gap by predicting serum hormone concentration over time from the user's logged dose history. It implements a pharmacokinetic model that computes, for any point in time, the superposed contribution of every dose the user has taken, plus a dynamic endogenous baseline, scaled by an optional per-user calibration factor derived from blood test results.

The model covers ten medication/route/form combinations across two hormone classes (Table 1). Its mathematical core is the Bateman biexponential—the analytic solution to one-compartment first-order absorption and elimination—which the published pharmacometric literature independently identifies as the best-fit model for every injectable testosterone ester examined: cypionate [1], enanthate [2], and undecanoate [5]. This convergence is not incidental; it reflects the fact that depot testosterone esters, across a wide range of chain lengths and dosing intervals, produce terminal phases that are single-exponential on semi-log plots, justifying one-compartment fits.

This paper describes the model in full: its governing equation, the provenance and derivation of every constant, the calibration mechanism, the accuracy of its predictions against published data, and the structural limitations that bound what it can honestly claim. It is written to be checkable: every constant is stated with its source, and every derivation is given in enough detail to reproduce.

1.1 Contributions

1. A complete description of a pharmacokinetic model designed for consumer HRT tracking, including the full constant set, per-form derivations, and the dynamic endogenous baseline model.
2. A conservation-of-mass identity (Section 2.8) that constrains the scaling constant S for all injectable testosterone esters to depend only on k_e and clearance, eliminating S as a free parameter and enforcing mutual consistency across esters.
3. An analytic closed-form derivation of S for testosterone cypionate from primary population-PK parameters (Section 2.7), which serves as the anchor for the identity above.
4. A per-user calibration framework ($\alpha = C_{\text{measured}}/C_{\text{predicted}}$) that closes part of the between-subject variability gap without requiring a formal mixed-effects model.
5. An honest accounting of structural limitations, including a source-conflict diagnosis for testosterone undecanoate's peak/trough ratio that reframes a previously assumed structural deficiency as a calibration problem.

2 Methods

2.1 System overview

Isocline predicts serum hormone concentration over time from a user's logged dose history. Two medication classes are supported, each across three administration routes, yielding ten medication/route/form combinations (Table 1).

Medication	Route	Ester / form	Output unit
Estradiol	Injectable	Valerate (EV), Cypionate (EC)	pg/mL
Estradiol	Oral	(route pseudo-ester)	pg/mL
Estradiol	Cream	(route pseudo-ester)	pg/mL
Testosterone	Injectable	Cypionate (TC), Enanthate (TE), Propionate (TP), Undecanoate (TU)	ng/dL
Testosterone	Oral	undecanoate (route pseudo-ester)	ng/dL
Testosterone	Cream	unesterified gel (route pseudo-ester)	ng/dL

Table 1: The ten medication/route/form combinations the model covers.

Route pseudo-esters. The two non-injectable cases in each ester enumeration describe a *route*, not a chemical ester. They are resolved from the dose record's administration method by a route-to-ester mapping function, never chosen by the user directly. This design ensures that a dose logged as "oral testosterone" uses the oral route's bioavailability and rate constants, not those of whatever injectable ester was previously selected.

Exogenous and endogenous separation. The PK engine returns the *exogenous* contribution only—the sum of all administered dose contributions. Endogenous production is modeled separately as a dynamic baseline $C_{\text{base}}(t)$ that decays as gonadal suppression progresses (Section 2.14). This separation matters for interpreting several constants, most importantly testosterone cypionate's half-life: the value encoded (3.84 d) is the exogenous-only disposition rate, not the total-testosterone terminal half-life of $\sim$7 d that is commonly quoted and that includes endogenous secretion.

2.2 Governing equation

Each administered dose contributes a Bateman biexponential—first-order absorption from a depot into a single well-mixed compartment, first-order elimination from it:

$$C_i(t) = D_i \cdot \text{EDF} \cdot S \cdot \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right), \qquad t \geq 0 \tag{1}$$

where D_i is the administered dose in mg, EDF is the effective dose factor (Section 2.5), S is a scaling constant with units of concentration per mg, and k_a, k_e are rate constants in day^{-1}. Total predicted level at time t is the superposition of all doses taken before it, plus the endogenous baseline, times a user calibration factor:

$$C(t) = \alpha \left[C_{\text{base}}(t) + \sum_{i\,:\,t_i \leq t} C_i(t - t_i) \right] \tag{2}$$

Superposition is exact here because Equation 1 is linear in dose. That linearity is also the model's principal simplification: there is no saturable protein binding, no SHBG dynamics, no dose-dependent clearance.

The $k_a \approx k_e$ limit. When $|k_a - k_e| < 10^{-4}$, Equation 1 is singular, and the engine substitutes its limit,

$$D \cdot \text{EDF} \cdot S \cdot k_a\, t\, e^{-k_a t},$$

which is the standard degenerate-case form of the Bateman function when the two rate constants coincide.

Transdermal route. The cream/gel route does not use Equation 1. It is modeled as 24 hourly micro-doses of $D/24$ each, decaying single-exponentially ($k_e = 0$), approximating a 24-hour square-wave release:

$$C_{\text{transdermal}}(t) = \sum_{h=0}^{23} \frac{D}{24} \cdot S \cdot e^{-k_a(t-t_h)}, \qquad t \geq t_h \tag{3}$$

where t_h is the time of the h-th micro-dose. This construction has no absorption phase; Section 5.1 explains why it under-models intraday variation.

2.3 Time to peak

Differentiating Equation 1 with respect to t and setting the derivative to zero gives the time to peak concentration:

$$t_{\max} = \frac{\ln(k_a/k_e)}{k_a - k_e} \tag{4}$$

A property of Equation 4 that has practical consequences is its *symmetry* under $k_a \leftrightarrow k_e$: swapping the two rate constants leaves $t_{\max}$ unchanged. For depot esters the two exponentials are not separately identifiable from clinical data, so which one is labelled "absorption" and which "elimination" is a modelling convention rather than a finding. The observable curve is unchanged by the swap, with the amplitude difference absorbed into S.

2.4 Flip-flop kinetics

For depot esters, absorption from the oil vehicle is slower than elimination of the released hormone ($k_a \ll k_e$). The terminal slope of the curve is then governed by *absorption*, not elimination— "flip-flop" kinetics [12]. The practical consequence is that the parameter controlling the terminal half-life sets k_a and governs the entire inter-dose peak-to-trough decay, while the parameter controlling the fast exponential sets k_e and primarily determines where the peak falls.

In Isocline's parameterization, the field `halfLifeDays` sets $k_a = \ln 2 / t_{1/2,\text{term}}$ (the slow rate), and `eliminationHalfLifeHours` sets $k_e = \ln 2 / (t_{1/2,\text{elim}}/24)$ (the fast rate). The naming is historical; the doc comments in the codebase explicitly note the convention.

2.5 Effective dose factor

$\text{EDF} = \text{EWF} \times F$ converts administered mg of ester into systemically available mg of active hormone.

2.5.1 Ester weight factor

The ester weight factor is pure stoichiometry—the mass fraction of the ester molecule that is the parent hormone:

$$\text{EWF} = \frac{M_{\text{hormone}}}{M_{\text{ester}}}$$

with $M_{\text{E}_2} = 272.4$ and $M_{\text{T}} = 288.4$ g/mol. These are the only constants in the entire model that are not fitted to anything and cannot be wrong. Table 2 gives the values for all esters.

Ester	Abbrev.	EWF	Molecular basis
Estradiol valerate	EV	0.764	$272.4/356.5$
Estradiol cypionate	EC	0.687	$272.4/396.6$
Estradiol (oral/TD)	E2	1.000	base hormone, no ester
Testosterone cypionate	TC	0.699	$288.4/412.6$
Testosterone enanthate	TE	0.720	$288.4/400.6$
Testosterone propionate	TP	0.837	$288.4/344.5$
Testosterone undecanoate	TU	0.631	$288.4/456.7$
Testosterone (gel)	T	1.000	base hormone, no ester

Table 2: Ester weight factors from molecular weights. Oral estradiol and transdermal routes deliver the base hormone directly, so EWF = 1.

Route	F	Basis
Injectable depot	1.0	Near-complete absorption from oil depot
Oral estradiol	0.05	First-pass metabolism
Transdermal E2	0.15	Skin barrier limits absorption
Transdermal T	0.10	AndroGel 1% label §12.3 (stated verbatim)
Oral TU	1.0	Not established; folded into S (see text)

Table 3: Route bioavailability values.

2.5.2 Bioavailability

F is route bioavailability—the fraction of the administered dose reaching systemic circulation:

Oral testosterone undecanoate carries $F = 1.0$ not because it is fully bioavailable—it is not—but because its absolute bioavailability is not established in the source. Since F and S are confounded (only their product is identified by the published data), folding bioavailability into S avoids inventing a number. Predictions are unaffected by where the split is drawn.

2.6 The scaling constant

S is an empirical surrogate for F/V_d. It is the one constant with no direct physical interpretation, and it is the constant most sensitive to calibration errors. Wherever the sources allow, S is derived in closed form rather than by curve fit; the derivations are given in Sections 2.7 and 2.8.

2.7 Testosterone cypionate: primary-source derivation

Testosterone cypionate (TC) is the only form in the model derived from a primary population-PK publication: Bi et al. [1], who report $\mathrm{CL}/F = 2.6\,\mathrm{kL/day}$, $V/F = 14.4\,\mathrm{kL}$, and $K_a = 1.22\,\mathrm{day}^{-1}$ (Table 2 of the paper, with reported relative standard errors and bootstrap confidence intervals).

2.7.1 Rate constants

The disposition rate is $\mathrm{CL}/V = 2.6/14.4 = 0.1806\,\mathrm{day}^{-1}$, giving a half-life of 3.839 d, which sets k_a (the slow rate under flip-flop). $K_a = 1.22$ gives $\ln 2/1.22 = 13.64\,\mathrm{h}$, which sets k_e (the fast rate). Per the symmetry noted after Equation 4, the labels are conventional; the observable curve is unchanged by the swap, with the amplitude difference absorbed into S.

2.7.2 Why 3.84 days and not 7–8

The commonly quoted TC half-life is around a week. Bi et al. address this discrepancy directly: noncompartmental analysis of their own data yields 6.9 d, and their model yields 6.87 d *when endogenous secretion is assumed to be zero*—but 4.05 d once endogenous production is accounted for. Because Isocline models endogenous production separately as $C_{\text{base}}(t)$, the exogenous-only figure is the one that belongs in the ester constant. Bi's own median post-hoc estimate of 4.05 d agrees with the encoded 3.839 d to 5.2%.

2.7.3 Scaling constant in closed form

Matching Equation 1 to the one-compartment model underlying Bi's parameters gives S analytically rather than by curve fit:

$$S = \frac{10^5 \cdot K_a}{V_L \cdot \text{EDF} \cdot k_e} = \frac{10^5 \cdot 1.22}{14400 \times 0.699 \times 0.180556} = 67.13 \tag{5}$$

The 10^5 converts mg/L to ng/dL; EDF appears because the app multiplies the dose by it while Bi's V/F already absorbs ester conversion. This reproduces Bi's population model exactly.

2.7.4 Time to peak

Bi's parameters give $t_{\max} = 1.84$ d via Equation 4. Propagating the published 95% CI on K_a (0.785–2.67) gives a band of 1.08–2.43 d. Bi's own limitations section (1–2 absorption-phase samples per subject, K_a RSE 44.9%) explains the imprecision. Read TC $t_{\max}$ as 1.84 ± 0.7 d.

2.8 Clearance identity for TE, TP, and TU

Testosterone enanthate (TE), propionate (TP), and undecanoate (TU) lack primary population-PK sources with reported V/F and K_a for the intramuscular route in the model's target population. Their scaling constants are nonetheless *not* free parameters, because of a conservation argument that follows from the TC derivation.

2.8.1 Derivation

Integrating Equation 1 over all time gives exactly

$$\int_0^\infty \frac{k_a}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right)\, dt = \frac{k_a}{k_a - k_e}\left(\frac{1}{k_e} - \frac{1}{k_a}\right) = \frac{1}{k_e}$$

so at steady state with dosing interval τ, superposition gives

$$C_{\text{avg}} = \frac{D \cdot \text{EDF} \cdot S}{\tau\, k_e} \tag{6}$$

The true steady-state average is (mg of testosterone per day) divided by clearance—and clearance is a property of *testosterone*, not of the ester carrying it. It is therefore already pinned by the TC refit: $\text{CL}_T = 2600 \times \text{EWF}_{TC} = 1817.4$ L/day. Equating the two expressions, the ester weight factor cancels:

$$\boxed{S = \frac{10^5 \cdot k_e}{\text{CL}_T \cdot F}} \tag{7}$$

S depends only on k_e and clearance. The ester's half-life (k_a) does not appear at all—a property worth stating plainly, because it means S does not track the ester's terminal half-life but rather its fast rate constant, which for depot esters under flip-flop is the absorption rate from the oil vehicle.

2.8.2 The load-bearing check

TC satisfies Equation 7 to 0.03% *without having been fit to it*—its S came from Equation 5, via V/F and K_a. Two independent derivations landing on the same number is what makes it defensible to apply the identity to the other three esters.

2.8.3 Application to TE, TP, and TU

Table 4 shows the identity applied. TE and TU now carry real absorption rates from primary sources (Section 2.9), while TP retains a placeholder 2 h k_e and is flagged as uncalibrated.

Ester	k_e (d^{-1})	S (encoded)	S (implied)	**Error**
Cypionate	1.2196	67.13	67.11	0.03%
Enanthate	1.9625	107.98	107.98	0.00%
Propionate	8.3178	457.67	457.67	0.00%
Undecanoate	0.3748	20.62	20.62	0.00%

Table 4: The clearance identity (Equation 7) applied to all injectable testosterone esters. $\mathrm{CL}_T = 1817.4$ L/day, $F = 1.0$ for all.

2.8.4 Steady-state predictions at typical regimens

Ester	**Regimen**	C_{avg}	$C_{\max}$	$C_{\min}$
TC	100 mg q7d	550	736	321
TE	100 mg q7d	566	703	405
TP	50 mg q2d	1151	1743	580
TU	1000 mg q70d	496	1045	129

Table 5: Steady-state predictions (exogenous only, ng/dL) at typical regimens. $C_{\max}$ and $C_{\min}$ are the peak and trough of the steady-state oscillation. Each row was recomputed for this revision both by forward simulation to steady state and, independently, by summing the analytic tail $\sum_{k\geq 1} C_i(k\tau)$; the two agree to four significant figures. The TP trough previously printed here (760 ng/dL) did not reproduce under either method and has been corrected to 580, which moves TP's steady-state peak/trough ratio from $2.29\times$ to $3.00\times$.

2.9 Testosterone enanthate and undecanoate: absorption rates

2.9.1 Testosterone enanthate (TE)

TE's absorption rate is derived from Ichihara et al. [2], the only identified primary IM pharmacokinetic study in transgender men: $n = 10$, single 250 mg IM dose (Enarmon Depot). The paper reports per-subject absorption rate constants (k_a) in the 4 of 10 subjects where it was estimable: 2.82, 3.16, 0.49, 1.38 day^{-1}, giving a mean of 1.9625 day^{-1} ($t_{1/2} = 8.48$ h). The terminal half-life is reported as 6.6 ± 1.9 d, which sets k_a (the slow rate) at $\ln 2/6.6 = 0.1050$ day^{-1}.

The resulting $t_{\max} = 1.58$ d agrees with the paper's observed 1.7 ± 0.9 d (all subjects peaked within 3 days). The $t_{\max}$ and the k_a values were estimated in different subsets (Tmax in all 10, k_a in the 4 where it was estimable), so this is a genuine cross-check rather than a round trip.

The mean of the estimable four is a censored mean. Ichihara's six remaining subjects were not missing at random: their absorption rate constants were reported as exceeding

10 day^{-1}, which is precisely why they could not be estimated from the available sampling and were instead analysed under an intravenous model. Averaging only the four estimable values—all of which fall below 3.2 day^{-1}—selects the slow tail of the distribution and biases the mean downward by an amount the paper does not let us quantify. Had the censored subjects been included at anything near their reported lower bound, the population mean would be several times 1.96 day^{-1}.

We keep 1.9625 day^{-1} anyway, for a reason worth stating rather than burying: a k_a of 10 day^{-1} paired with the reported 6.6 d terminal half-life puts $t_{\max}$ at 0.46 d, and no subject in the study peaked that early—the observed mean was 1.7 ± 0.9 d across all ten. The censored subjects' fitted rates and the cohort's own observed $t_{\max}$ cannot both be right. This is a source-internal conflict of the same species as the TU one (Section 5.1.1), and the constant is chosen to match the observable ($t_{\max}$) rather than the fitted parameter. Note also that S is pinned to k_e by the clearance identity, so a future revision of k_a must move S with it; total exposure ($\mathrm{AUC} = D^{\mathrm{eff}} S / k_e$) is invariant under that pair, and only the curve's shape is at stake.

Magnitude discrepancy. A single 250 mg dose gives 881 ng/dL exogenous here against Ichihara's observed $C_{\max}$ of 2940 ± 1710 ng/dL. The gap is cohort-level, not error: Ichihara's own $\mathrm{CL_{tot}}/F$ (14.3 L/day/kg $\approx$ 930 L/day at 65 kg) is about half the $\mathrm{CL}_T = 1817.4$ L/day inherited from Bi, and their V_d/F (118.7 L/kg $\approx$ 7700 L) about half Bi's 14 400 L. Adopting Ichihara's clearance for TE would put TE and TC $2\times$ apart on the *same molecule*, breaking the conservation-of-mass identity. Magnitude is therefore anchored to Bi; only shape was taken from Ichihara. The per-user calibration factor α exists to absorb individual-level differences of this magnitude, and Ichihara's $C_{\max}$ CV of 0.58 confirms that between-subject spread is wide enough to accommodate the gap.

2.9.2 Testosterone undecanoate (TU)

TU's terminal half-life (20.0 d) is retained from Zhang et al. [3], who measured 18.3 ± 2.3 d (500 mg) and 23.7 ± 2.7 d (1000 mg); 20.0 sits inside both arms. The absorption rate is fitted so that $t_{\max}$ lands on the Aveed (FDA) clinical pharmacology median of 7 d (range 4–42 d): solving Equation 4 with $k_a = \ln 2 / 20.0 = 0.0347$ day^{-1} for k_e gives $k_e = 0.3748$ day^{-1} ($t_{1/2} = 44.4$ h), and $t_{\max} = 7.00$ d.

The scaling constant follows from the identity: $S = 10^5 \times 0.3748 / 1817.4 = 20.62$.

Peak/trough ratio. Over a 70-day interval, the model produces a peak/trough ratio of 8.1$\times$ (1045/129 ng/dL). Wang et al. [4] report a steady-state peak/trough of 1.8$\times$ (789.8/435.6 ng/dL at week 12). This gap is *arithmetic, not structural*: over a 70-day interval, a terminal half-life of 20 d is 3.5 half-lives of decay, so peak/trough cannot fall below $\sim$11$\times$ on decay alone. Both targets *are* jointly reachable in the two-rate Bateman—a 73.8 d terminal half-life paired with a 27.5 h release rate hits $t_{\max} = 7$ d *and* peak/trough 1.80$\times$ simultaneously—but 73.8 d is 3.7$\times$ what Zhang measured. The blocker is that Zhang's measured terminal half-life and Wang's observed flatness cannot both be true; resolving it requires Wang's primary sampling schedule, not a model rewrite. This finding is consistent with Pastuszak et al. [5], who fit injectable TU with a one-compartment first-order-absorption model—the same two-rate-constant Bateman used here.

2.9.3 Testosterone propionate (TP)

TP remains uncalibrated. It retains a placeholder 2 h k_e (physiological clearance of free testosterone), which pins its $t_{\max}$ to 0.33 d against a published window of 1–2 d. Its *magnitude* was corrected by the clearance identity (Section 2.8), so steady-state C_{avg} is correct, but the

first ∼24 h of the curve—where $t_{\max}$ lives—is wrong. The best available source is a textbook chapter [10] showing a simulated twice-weekly curve rather than fitted rate constants or raw single-dose data. A fabricated fit would be worse than the honest flag.

2.10 Estradiol ester derivations

2.10.1 Estradiol valerate (EV) and cypionate (EC)

EV and EC are calibrated against transfemscience.org's compiled tables [8], which aggregate published single-dose and steady-state estradiol PK. The calibration procedure:

1. k_e was fit so Equation 4 reproduces the published time-to-peak (2.1 d for EV, 3.1–4.3 d for EC).
2. k_a was fit so the modeled steady-state trough reproduces Table 10's 5 mg-weekly figures.
3. S was anchored to the same table's steady-state peak.

The resulting constants reproduce the published steady-state $C_{\max}$ and $C_{\min}$ to within 0.1% (Table 7). A key decision: single-dose and steady-state targets were *not simultaneously satisfiable* in the two-parameter model, and steady state was chosen on the reasoning that it is what users actually live at and what clinicians titrate on. The consequence is that single-dose $C_{\max}$ is a derived value, not an enforced target (Section 5.2).

2.10.2 Oral estradiol

Oral estradiol uses the same Bateman biexponential with fast rate constants. The absorption rate $k_a = 8.762\,\text{day}^{-1}$ ($t_{1/2} = 0.0791$ d) was fit so that $t_{\max} = 6.0$ h, the midpoint of the published 4–8 h window. The elimination rate $k_e = 1.386\,\text{day}^{-1}$ ($t_{1/2} = 12.0$ h) and $S = 1250$ were anchored to reproduce the steady-state average for 2 mg daily (90 pg/mL).

A coupling issue arises: S and k_a are coupled through the peak but not through the average (integrating the Bateman term gives $\text{AUC/dose} = D^{\text{eff}} S/k_e$, with no k_a in it). One (S, k_a) pair cannot hold both the published $t_{\max}$ and the published single-dose $C_{\max}$. Anchoring on C_{avg} keeps 2 mg daily at 90 pg/mL—unchanged by the k_a refit, and the quantity clinicians titrate on—at the cost of single-dose $C_{\max}$ reading 88 pg/mL rather than 60 pg/mL for 2 mg.

2.10.3 Transdermal estradiol

Transdermal estradiol uses the 24 hourly micro-dose construction (Equation 3) with $S = 155$, $F = 0.15$, and $k_a = \ln 2/1.5 = 0.462\,\text{day}^{-1}$. The stored k_e is unused because the micro-dose branch sets $k_e = 0$ (single-exponential decay).

2.11 Transdermal testosterone

Anchored to the AndroGel 1% prescribing information [7], Table 5 (Day 180 steady state, $n = 44$): 50 mg/day → $C_{\text{avg}} = 555$ ng/dL. The terminal half-life comes from the label's washout statement—levels return to baseline in 48–72 h, implying ∼5% residual at 60 h, giving $t_{1/2} = 0.5784$ d. Transdermal $F = 0.10$ is stated outright in the label's §12.3: "Approximately 10% of the testosterone dose applied on the skin surface from AndroGel is absorbed into systemic circulation."

The scaling constant $S = 132.98$ is derived from the 50 mg anchor. Two limitations are documented rather than fitted around; both are in Section 5.1.

2.12 Oral testosterone undecanoate

Anchored to Swerdloff et al. [6] / the Jatenzo label: 237 mg twice daily → $C_{\text{avg}} = 488.7$ ng/dL, $t_{\max}$ 2–4 h. Both rate constants were solved jointly so that $t_{\max}$ lands at 3 h and the steady-state $C_{\max}/C_{\text{avg}}$ comes as close as the model permits to the reported 2.06. The resulting

constants are $k_a = 7.702\,\text{day}^{-1}$ ($t_{1/2} = 0.09\,\text{d}$) and $k_e = 8.305\,\text{day}^{-1}$ ($t_{1/2} = 2.003\,\text{h}$), with $S = 13.57$.

2.13 The basal offset used in the testosterone comparisons

Several testosterone rows in Sections 4 and 4.4 report a total level rather than an exogenous one, formed by adding a flat 240 ng/dL basal to the model's exogenous output. That figure deserves a name, because it is not something the model computes.

It is the residual endogenous testosterone of Bi et al.'s cohort: healthy eugonadal men whose own gonadal production is partially, not completely, suppressed by exogenous cypionate. It is the right offset for comparing model output against Bi's *total*-testosterone observations, which is the only place it is load-bearing (Table 8).

It is *not* what the app's own baseline model (Section 2.14) produces for any user at any time. That model runs from 600 ng/dL to a suppressed floor and passes through 240 only transiently, roughly 1.5 weeks into the taper. Where the persona table (Section 4.4) carries the same 240 offset, it should be read as "the Bi-cohort comparison basis," not as the level the application would display for that persona. Section 5.1.6 takes up the more serious problem this exposes.

2.14 Dynamic endogenous baseline

Endogenous hormone production is modeled as a time-varying baseline $C_{\text{base}}(t)$ that decays from an intact gonadal level to a suppressed level as HRT progresses. The model uses a saturating exponential taper:

$$C_{\text{base}}(t) = C_{\text{full}} + (C_{\text{supp}} - C_{\text{full}}) \cdot \left(1 - e^{-2.5\,p(t)}\right) \tag{8}$$

where $p(t) = \min(\text{weeks on HRT}/t_{\text{scale}}, 1)$ is the normalized progress, t_{scale} is user-configurable (4, 8, or 12 weeks), C_{full} is the intact baseline (100 pg/mL for estradiol, 600 ng/dL for testosterone), and C_{supp} is the suppressed baseline (5 pg/mL for estradiol, 15 ng/dL for testosterone). The factor 2.5 in the exponent ensures the suppression reaches $\sim$92% at $p = 1$. After 104 weeks (2 years), the baseline is treated as fully suppressed.

The 92% plateau is a discontinuity, not an asymptote. Because $p(t)$ is clamped at 1, Equation 8 stops decaying once t_{scale} is reached and holds at $1 - e^{-2.5} = 91.8\%$ of the way down—63 ng/dL for testosterone, 12.8 pg/mL for estradiol—for the remainder of the first two years. The 104-week rule then drops it to the suppressed value in a single step. A user therefore carries a fixed 48 ng/dL (or 7.8 pg/mL) of modeled endogenous hormone for two years and loses it overnight on the anniversary. Nothing physiological corresponds to that step. Removing the clamp on $p(t)$—letting the exponential continue to converge—makes the taper continuous and renders the 104-week rule a no-op, since $e^{-2.5 \cdot 26} \approx 0$. This is a defect in the implementation, recorded here rather than silently smoothed in the figures.

Anti-androgen override. When the user reports taking an anti-androgen (spironolactone, cyproterone, or a GnRH agonist), the testosterone baseline is set to the suppressed level immediately, reflecting the direct gonadal suppression these agents provide.

2.15 Blood test calibration

2.15.1 The calibration factor

Each blood test record stores a measured hormone level and the date the blood was drawn. The calibration factor is the ratio of measured to predicted concentration at the draw instant:

$$\alpha = \frac{C_{\text{measured}}}{C_{\text{predicted}}} \tag{9}$$

where $C_{\text{predicted}}$ is computed by running the PK model at the exact draw date *without* calibration applied (to avoid circularity: the calibration engine must never feed on its own output). The factor is clamped to $[0.2, 5.0]$ to exclude clinically implausible values. When multiple calibrated blood tests exist, the most recent one for the currently selected hormone type is used.

2.15.2 Application, and a round-trip defect

Equation 2 as written applies α to the entire model output, baseline included. The implementation does not: it computes `baseLevel + cumulativeSpike * alpha`, scaling only the exogenous sum. The two disagree, and the disagreement breaks the definition of α in Equation 9.

α is derived as $C_{\text{measured}}/C_{\text{predicted}}$ where $C_{\text{predicted}} = C_{\text{base}} + C_{\text{exo}}$, but is then applied to C_{exo} alone. Re-evaluating the model at the draw instant therefore returns

$$C_{\text{base}} + \alpha\, C_{\text{exo}} = C_{\text{measured}} + (1 - \alpha)\, C_{\text{base}}$$

rather than C_{measured}. The residual $(1 - \alpha)C_{\text{base}}$ is zero only when $\alpha = 1$ or the baseline is negligible. It is negligible for a fully suppressed user ($C_{\text{base}} = 5$ pg/mL or 15 ng/dL, giving well under 1%), and it is not negligible early in therapy, where the baseline is largest: at $C_{\text{base}} = 328$ ng/dL and $C_{\text{exo}} = 400$ ng/dL, a measured 500 ng/dL yields $\alpha = 0.687$ and a post-calibration prediction of 603 ng/dL—a $+20.5\%$ error at the very point the calibration was supposed to pin down. The error is always in the direction that resists the correction: a user whose true levels run below the population model stays over-predicted.

Of the two readings, scaling C_{exo} alone is the physiologically defensible one—α corrects the drug model, and the user's adrenal and gonadal output is not a function of their injection technique or depot absorption. The fix is therefore to the derivation rather than to the application:

$$\alpha = \text{clamp}\left(\frac{C_{\text{measured}} - C_{\text{base}}(t_{\text{draw}})}{C_{\text{exo}}(t_{\text{draw}})},\ 0.2,\ 5.0\right) \tag{10}$$

which round-trips exactly, and which requires a guard for $C_{\text{measured}} \le C_{\text{base}}$—a draw at or below the modeled endogenous floor, where no positive α can reconcile the two and the honest response is to decline to calibrate rather than to clamp to 0.2. Equation 2 is retained above as the intended semantics; Equation 10 is what makes it true.

Subject to that correction, α is a pragmatic per-user analog of the inter-individual variability (IIV) terms in a formal mixed-effects population-PK model. The published literature reports substantial between-subject variability: Ichihara et al. [2] report a CV of 0.58 on V_d/F for TE, corresponding to an $\sim$8-fold individual range. The clamp range $[0.2, 5.0]$ spans most of this distribution without admitting values that would indicate a data entry error.

2.15.3 Hormone-type gating

The calibration factor is only applied when the blood test's hormone type matches the currently selected medication. This prevents a testosterone draw from silently rescaling the estradiol curve, which would occur if the filter were absent.

2.16 Drug interactions: spironolactone

The application offers an opt-in adjustment, off by default, that scales k_e for estradiol doses by 1.15 when the user reports taking spironolactone. It was introduced on the reasoning that spironolactone induces CYP3A4 and thereby modestly accelerates estradiol clearance. That reasoning does not survive review, and this section is rewritten to say so plainly.

$$k_e' = k_e \cdot 1.15 \tag{11}$$

This adjustment is applied at the PK parameter resolution level, before the Bateman function is evaluated. It is not applied to testosterone doses. Because $t_{\max}$ depends on k_e (Equation 4), it also shifts the modeled peak slightly earlier. The ester-level $t_{\max}$ values in Table 9 are computed without it.

The magnitude is unsourced. No primary human study establishes a 15% increase in estradiol clearance under spironolactone. The 1.15 is a round number chosen from the midpoint of an informal "10–20%" range that this paper previously attributed to "typical estimates" without a citation, and none can be supplied. The one published study that measured estradiol pharmacokinetics directly in transfeminine people taking spironolactone—Cattani et al. [9], in Brazilian transgender women—reports estradiol exposure unchanged by the concomitant medication it was testing (AUC geometric mean ratio 0.89, 90% CI 0.76–1.04; trough ratio 1.06, 90% CI 0.94–1.20) and identifies no spironolactone-driven effect on estradiol. The best available evidence is therefore consistent with no interaction of the modeled size.

The mechanism is applied to the wrong exponential. Under the flip-flop convention this model adopts (Section 2.4), k_e for a depot ester is the *fast* rate, which for an oil depot is absorption from the vehicle—not hepatic clearance. Clearance governs the slow exponential, which the app stores in `halfLifeDays`. Scaling k_e does reduce exposure in roughly the right direction, because $\mathrm{AUC}/\mathrm{dose} = D^{\mathrm{eff}} S / k_e$ and S is held fixed, but it does so by making the depot release faster while leaving the terminal slope untouched. A genuine clearance increase would steepen the terminal decay and deepen the trough; this adjustment does neither. The effect on the curve's *shape* is therefore wrong even where its effect on total exposure is arguably right, and the two cannot be separated in the current parameterization.

Disposition. The adjustment is retained, because removing a user-facing toggle is a product decision rather than a modelling one, but it is demoted to the bottom tier of the source hierarchy (Section 5.3) alongside testosterone propionate, and it should not be presented in the interface as a fact about spironolactone. It is an opt-in sensitivity knob whose default—off—is the position the evidence supports.

2.17 Aromatization estimation

For trans men on testosterone therapy, a small fraction of circulating testosterone is converted to estradiol via aromatase. The model estimates this conversion as:

$$E_2^{\mathrm{arom}} = f_{\mathrm{arom}} \cdot 10 \cdot T_{\mathrm{ng/dL}} = 0.02 \times T_{\mathrm{ng/dL}} \quad \text{(pg/mL)} \tag{12}$$

where $f_{\mathrm{arom}} = 0.002$ is the modeled mass fraction of circulating testosterone converted to estradiol, and the factor of 10 is the unit conversion $1\ \mathrm{ng/dL} = 10\ \mathrm{pg/mL}$. Both hormones are expressed in mass-per-volume, so the fraction applies to concentrations directly; the factor of 10 carries ng/dL into pg/mL and is not optional.

Correction. Earlier revisions of this paper stated Equation 12 as $E_2 = 0.002 \times T_{\text{ng/dL}}$, omitting the unit conversion and understating the implemented estimate by a factor of ten. The implementation (`estimatedE2FromTestosterone`) has always carried the $\times 10$; the paper was wrong, not the code.

Sanity check. At a mid-cycle T of 600 ng/dL the estimate is 12 pg/mL, at the low end of the quoted male reference range (10–40 pg/mL); at 900 ng/dL it is 18 pg/mL. The serum E_2/T ratio in eugonadal men is nearer 0.5% on a mass basis, so $f_{\text{arom}} = 0.002$ is conservative by roughly a factor of two. It is retained rather than retuned because no primary source in this model's hierarchy fixes it, and because a low estimate fails safe for the use it is put to—telling a trans man whether his estradiol is running *high*. The conversion factor is a population average; individual aromatase activity varies with body composition, age, and genetic factors, and it is not a substitute for a measured E_2.

2.18 Chart window and sampling resolution

The chart's visible time range and sampling step are derived from the ester's pharmacokinetic properties, not hardcoded. The rule is:

$$W = \text{clamp}(1.5 \cdot t_{\max},\ \text{lower: } \min(2\,\text{d},\ 2\tau),\ \text{upper: } 1.5\tau) \tag{13}$$

$$\text{step} = \max\big(\min(W/40,\ t_{\max}/8),\ 15\,\text{min}\big) \tag{14}$$

where W is the total window width, τ is the dosing interval, and the window is split 1:2 past:future. The $t_{\max}/8$ cap is an anti-aliasing guarantee: it ensures the sampling step never exceeds one-eighth of the time-to-peak, preventing the drawn curve from collapsing into a flat line when the step would otherwise land on the same phase of every dose cycle. The $W/40$ term keeps every ester at a comparable point count ($\sim$41) so curves read with the same density regardless of timescale.

When $t_{\max}$ is uncalibrated or non-finite (the transdermal routes, which have no meaningful $t_{\max}$ because $k_e = 0$), a fallback window of 2 d is used.

3 Encoded constants

	slow exponential		fast				
Form	$t_{1/2}$ (d)	k_a (d^{-1})	k_e (d^{-1})	EWF	F	S	$t_{\max}$
EV	2.6000	0.2667	0.7737	0.764	1.00	396.40	2.10 d
EC	6.4700	0.1071	0.4310	0.687	1.00	273.00	4.30 d
E2 oral	0.0791	8.7629	1.3859	1.000	0.05	1250.00	6.0 h
E2 cream	1.5000	0.4621	—	1.000	0.15	155.00	—
TC	3.8390	0.1806	1.2196	0.699	1.00	67.13	1.84 d
TE	6.6000	0.1050	1.9625	0.720	1.00	107.98	1.58 d
TP	1.0000	0.6931	8.3178	0.837	1.00	457.67	0.33 d
TU	20.0000	0.0347	0.3748	0.631	1.00	20.62	7.00 d
T oral (TU)	0.0900	7.7016	8.3053	0.631	1.00	13.57	3.0 h
T gel	0.5784	1.1984	—	1.000	0.10	132.98	—

Table 6: Constants as shipped, and the $t_{\max}$ each implies via Equation 4. The cream/gel routes have no $t_{\max}$ because they are not modeled with Equation 1; their k_e is stored but unused.

Calibration status. Of the eight forms with a $t_{\max}$, seven are calibrated against a primary or label source: EV and EC against compiled steady-state tables [8], oral E2 against the published 4–8 h window, TC against Bi et al. [1], TE against Ichihara et al. [2], TU against Aveed's FDA clinical pharmacology, and oral T against the Jatenzo label [6]. TP alone remains uncalibrated: its $t_{\max}$ of 0.33 d is known to be wrong against a published window of 1–2 d, and is not presented to users as a fact about the drug.

4 Results

4.1 Accuracy against anchors

Regimen	Quantity	Model	Published	Error	Source
EV 5 mg q7d	$C_{\max}$	384	384	$+0.1\%$	TFS Table 10
EV 5 mg q7d	$C_{\min}$	142	142	$+0.1\%$	TFS Table 10
EC 5 mg q7d	$C_{\max}$	339	339	-0.1%	TFS Table 10
EC 5 mg q7d	$C_{\min}$	262	262	-0.1%	TFS Table 10
TC 100 mg q7d	C_{avg}	550	549	$+0.0\%$	Bi Table 2
T gel 50 mg/d	C_{avg}	555	555	$+0.1\%$	AndroGel Table 5
T gel 75 mg/d	C_{avg}	833	601	$+38.6\%$	AndroGel Table 5
T gel 100 mg/d	C_{avg}	1111	713	$+55.8\%$	AndroGel Table 5
Oral TU 237 mg BID	C_{avg}	489	489	-0.0%	Jatenzo
Oral TU 237 mg BID	$C_{\max}$	789	1008	-21.7%	Jatenzo

Table 7: Predicted versus published. Rows with near-zero error are the anchors themselves and demonstrate that the fit converged. The three coloured rows are the informative ones; both are explained in Section 5.1. Estradiol in pg/mL, testosterone in ng/dL.

Rows in Table 7 with 0.0% error demonstrate only that the fit converged—a fit agrees with its own target by construction. The three coloured rows are the informative ones.

4.2 Out-of-sample validation

The following comparisons were *not* used in any fit.

Check	Model	Published	Error	Source
TC single 200 mg $C_{\max}$ (exogenous)	997	872	$+14.3\%$	200 mg IM lit. 1112 less 240 basal
TC 100 mg q7d $C_{\max}$ + basal	976	1150	-15.1%	Bi Figure 2, 100 mg panel
EV single 2 mg $C_{\max}$	119	118	$+1.0\%$	TFS Table 9
EC single 2 mg $C_{\max}$	59	62	-5.1%	TFS Table 9

Table 8: Independent validation. The estradiol rows are the strongest evidence: constants fit purely to steady-state data reproduce single-dose peaks within 1% and 5%.

The two testosterone rows agree to about 15% in opposite directions, which is roughly the precision the source itself supports given Bi's reported RSEs. They are consistency checks, not tight validation.

The estradiol rows are the strongest evidence in the model: constants fit purely to steady-state data reproduce single-dose peaks within 1% and 5%. This is a non-trivial result—it demonstrates that the two-parameter Bateman, with steady-state-anchored constants, generalizes to the single-dose regime without refitting.

Form	Model $t_{\max}$	Published	Source quality
EV	2.10 d	2.1 d	compiled tables
EC	4.30 d	3.1–4.3 d	compiled tables
E2 oral	6.0 h	4–8 h	secondary
TC	1.84 d	1.84 d	**primary** (Bi Table 2)
TE	1.58 d	1.7 ± 0.9 d	**primary** (Ichihara 2020)
TP	**0.33 d**	**1–2 d**	sparse (uncalibrated)
TU	7.00 d	median 7 d	**label** (Aveed FDA)
T oral	3.0 h	2–4 h	label (Jatenzo)

Table 9: Where the model puts the peak. Seven of eight forms with a $t_{\max}$ are calibrated against a primary or label source. TP alone remains uncalibrated: its $t_{\max}$ is known to be wrong and is not presented to users as fact.

4.3 Time to peak

4.4 Persona validation

Section-level agreement with published tables is necessary but not sufficient: those tables describe clean steady state on a fixed regimen, and real users miss doses, change dose, change route, and dose on intervals nobody studied. Eight synthetic users were run end to end through the same summation the app performs.

Persona	Regimen	What it tests, and the result
Maya, 24	4 mg EV IM q7d, 18 months, perfect adherence	Control. $C_{\max}$ 307, $C_{\min}$ 114 pg/mL — exactly $4/5$ of the 5 mg source figures, as linearity requires. Pass.
Devon, 31	100 mg TC q7d $\rightarrow$ 200 mg q14d at month 6	Superposition across a regimen change. C_{avg} preserved at 790 ng/dL as it must be; trough falls $561 \rightarrow 381$ while peak rises $976 \rightarrow 1339$, and peak/trough widens $1.74\times \rightarrow 3.51\times$. The model reproduces the clinical trade-off and correctly refuses to reward the longer interval. Pass.
Priya, 47	2 mg oral E2 BID for 4 months, then 4 mg EV q7d	Route switch; historical doses must keep their own kinetics. They do — the oral segment retains $F = 0.05$ and the fast rates. Modeled BID swing is $1.37\times$ at an unchanged C_{avg} of 180 pg/mL. Pass.
Sam, 19	50 mg/day T gel, misses $\sim$2 days/week	Skipped-dose handling. C_{avg} 637 ng/dL; missed days cost 20% of average and 63% of trough. Pass.
Rowan, 38	4 mg/day compounded E2 cream, onboarded in-app	Route pseudo-ester resolution. Cream dose correctly resolves to the transdermal pseudo-ester constants. Pass.
Kai, 52	237 mg oral TU BID, 10-day interruption	Washout and re-accumulation on the fastest route. Falls to the endogenous floor within a day; no phantom drug survives the gap; recovers to 100% of pre-gap within 2 days. Pass.

Persona	Regimen	What it tests, and the result
Jules, 28	100 mg TE IM q7d	TE steady-state. $C_{\max}$ 703, $C_{\min}$ 405, C_{avg} 566 ng/dL (exogenous). Consistent with TC via the clearance identity. $t_{\max}$ 1.58 d, inside Ichihara's observed 1.7 ± 0.9 d. Pass.
Nadia, 35	5 mg EC q10d — an interval nobody studied	Extrapolation outside the calibration envelope. Trough holds in band but sits 40% below the same dose at q7d. Structurally sound but inherits every assumption of the fit. Pass, with caveat.

Basal offset. The testosterone personas above quote totals formed by adding the 240 ng/dL Bi-cohort basal (Section 2.13) to exogenous output. That offset is a comparison basis, not a level the application would display for a transmasculine user; see Section 5.1.6. The estradiol personas quote exogenous figures without an offset. Reading the two columns as like-for-like would be a mistake.

All eight land inside their clinical reference bands. The persona suite tests properties that section-level accuracy tables cannot: linear scaling (Maya), superposition across regimen changes (Devon), route switching with per-dose kinetics (Priya), skipped-dose handling (Sam), route pseudo-ester resolution (Rowan), washout and re-accumulation (Kai), inter-ester consistency (Jules), and extrapolation outside the calibration envelope (Nadia).

5 Discussion

5.1 Structural limitations

These are quantified rather than fixed. Each needs either a source the model does not have or a structurally different approach.

5.1.1 Testosterone undecanoate peak/trough ratio

Over a 70-day interval, the model produces a peak/trough ratio of 8.1× against Wang et al.'s [4] observed 1.8×. This is not a structural deficiency of the two-rate Bateman: as noted in Section 2.9, both targets are jointly reachable with a 73.8 d terminal half-life, but that value is 3.7× what Zhang et al. [3] measured. The blocker is a *source conflict* between Zhang's terminal half-life and Wang's observed flatness, not the number of compartments. Pastuszak et al. [5] fit injectable TU with the same one-compartment first-order-absorption model used here, confirming the model family is appropriate.

The magnitude correction via the clearance identity improves TU's C_{avg} (496 ng/dL exogenous, against a published ∼500–600 total) but its $C_{\max}$ overshoots (1045 vs. 790 ng/dL observed), because correcting magnitude cannot correct shape. The trade was taken knowingly: C_{avg} is what clinicians titrate on and what a user lives at.

5.1.2 Transdermal intraday ripple

The 24-hourly-micro-dose construction has no absorption phase and tops out at $C_{\max}/C_{\min} \approx 1.05$ for testosterone gel, against an observed 2.24. Reproducing 2.24 would require a terminal half-life near 1 h, which contradicts the label's own 48–72 h washout by an order of magnitude. The two cannot be satisfied together in this model. C_{avg} is exact.

5.1.3 Transdermal dose-response sublinearity

The AndroGel 1% label's dose-response is markedly sublinear: 50 mg → 555, 75 mg → 601, 100 mg → 713 ng/dL. The 1.62% label inverts outright: 81 mg C_{avg} 537 is *below* 60.75 mg C_{avg} 643.

This is a titration artifact: higher-dose arms are enriched for poor absorbers, because subjects were escalated precisely when they responded badly. It is selection confounding, not saturable pharmacokinetics, and a linear model should not be bent to reproduce it. S is anchored to the 50 mg arm—the standard starting dose and the least confounded—so predictions at 75 and 100 mg run high by 39% and 56% (Table 7).

Gel absorption also varies enormously between individuals. The calibration factor α is the intended remedy and is more important on this route than on any other.

5.1.4 Oral testosterone peak fluctuation

With $t_{\max}$ pinned to the reported 2–4 h window, the model's achievable $C_{\max}/C_{\text{avg}}$ tops out at 1.61 against a reported 2.06, so $C_{\max}$ reads 789 rather than 1008 ng/dL.

The reported ratio is itself inflated by mean-of-maxima bias: the published $C_{\max}$ is the mean of per-subject maxima, and $\mathbb{E}[\max] \geq \max[\mathbb{E}]$, with a reported SD of ± 581 on a mean of 1008. Anchoring to C_{avg}—the label's own primary endpoint—and accepting a documented under-prediction of the peak was judged the better trade.

5.1.5 Testosterone propionate remains uncalibrated

TP retains a placeholder 2 h k_e, which pins its $t_{\max}$ to 0.33 d against a published window of 1–2 d. Its magnitude was corrected by the clearance identity, so steady-state C_{avg} is correct, but the peak arrives too early and the curvature between peak and trough is wrong. The best available source is a textbook chapter [10] showing a simulated curve, not fitted rate constants. A fabricated fit would be worse than the honest flag.

5.1.6 The endogenous baseline is keyed to the wrong variable

This is the most consequential defect identified in this review, and unlike the others it is not a source conflict or an anchoring trade-off. It is a modelling error.

Equation 8 needs two numbers per hormone: the intact gonadal baseline C_{full} and the suppressed floor C_{supp}. The implementation supplies one pair per *hormone*—100 and 5 pg/mL for estradiol, 600 and 15 ng/dL for testosterone—and selects between them by asking which hormone is currently being tracked. But the intact gonadal baseline is not a property of the hormone being tracked. It is a property of the gonads the user has.

The two constants do not even describe the same person. A testosterone baseline of 600 ng/dL is the intact eugonadal male value; an estradiol baseline of 100 pg/mL is a mid-follicular value from an ovulatory cycle. One is an AMAB starting point, the other an AFAB one, and the model applies whichever the tracked hormone selects:

- A transfeminine user tracking estradiol is given 100 pg/mL of modeled endogenous estradiol at week 0. Her actual pre-HRT estradiol is on the order of 20–30 pg/mL. The model over-states her endogenous contribution by roughly fourfold.
- A transmasculine user tracking testosterone is given 600 ng/dL of modeled endogenous testosterone at week 0. His actual pre-T testosterone is on the order of 15–50 ng/dL. The model over-states it by more than an order of magnitude—and narrates the ensuing decay as "gonadal suppression," which is not what testosterone therapy does to him.
- A transfeminine user tracking *testosterone* to confirm suppression is served correctly: 600 decaying to 15 is exactly her case.

The mapping is therefore right for one of the three and wrong for the other two, and nothing in the data model distinguishes them.

Magnitude. At the default 4-week suppression timescale, the phantom testosterone baseline is 328 ng/dL at week 1, 183 at week 2, and 63 from week 4 onward (Section 2.14's plateau). Against a TC 100 mg q7d regimen whose exogenous contribution is still accumulating over that same fortnight, this roughly doubles the predicted level during the first two weeks of therapy—the period in which a new user is most likely to be checking the number and least likely to have a blood test to contradict it. The estradiol error is smaller in ratio but lands on a smaller quantity: 92 pg/mL of phantom baseline at week 1, against a 4 mg EV dose whose own peak is 238 pg/mL.

And the baseline is flat in time. A third, smaller problem sits underneath both. `estimatedLevels` evaluates Equation 8 *once*, as of the moment the series is built, and adds that single value to every point in it—including points months in the past. The endogenous floor drawn beneath a six-month chart is therefore a horizontal line at today's suppressed value, not the decaying curve Equation 8 describes. A user's first months are drawn as though they were already suppressed. The same flattening reaches the calibration engine, which compares a lab drawn in month 1 against a prediction that assumes month-1 suppression equalled today's. Evaluating the baseline per sample point resolves it, at the cost of re-scaling every stored α—a migration, not an edit.

It compounds with the calibration defect. A blood test is the mechanism by which a user is supposed to correct exactly this kind of population-level error. But α is derived against $C_{\text{base}} + C_{\text{exo}}$ and applied to C_{exo} alone (Section 2.15), so an inflated baseline both raises the prediction and blocks the correction from reaching it. The two defects are worst in the same regime—early therapy, large baseline—and they push the same direction.

What a fix requires. It requires a signal the model does not currently carry: the user's gonadal starting point, or equivalently the direction of therapy. The tracked hormone is not a proxy for it, as the third case above shows. Adding an explicit therapy-direction field and keying C_{full} to *that* would resolve all three cases with no change to Equation 8 itself; it is a data-model change, not a pharmacokinetic one. Until it exists, this paper's testosterone comparisons quote exogenous-only figures with the basal offset stated explicitly (Section 2.13), which is the most honest presentation available.

5.1.7 Model-wide simplifications

The model assumes:

- **Dose-linearity** with no saturable protein binding or SHBG dynamics.
- **A single well-mixed compartment** with constant clearance.
- **No covariates**: body composition, age, hepatic function, and concomitant medications (beyond the one spironolactone adjustment) are not represented.
- **No between-subject variability** in the population prediction. Predictions are population means, not individual forecasts. The per-user calibration factor α exists to close some of that gap once blood tests are available.

These are standard simplifications for a one-compartment model and are shared by the published population-PK models cited above [1, 2, 5], with the caveat that those models additionally include IIV terms and covariates (e.g., body weight and SHBG in Pastuszak's TU model).

5.2 The steady-state vs. single-dose anchor choice

For both estradiol esters and all four testosterone esters, the single-dose peak targets and the published steady-state data were *mutually unsatisfiable* in the two-parameter model. For

estradiol the gap was large: holding the single-dose targets put steady state $+110\%$ (EV) and $+206\%$ (EC) above the published table.

Steady state was chosen, in every case, on the reasoning that it is what users actually live at and what clinicians titrate on. The consequence is that single-dose figures are derived values, not enforced targets. This is a modelling decision, not a derivation, and it has a cost: single-dose $C_{\max}$ for EV (119 pg/mL for 2 mg) and EC (59 pg/mL for 2 mg) are lower than some published single-dose figures, though they agree with the transfemscience Table 9 values to within 1% and 5% respectively (Table 8).

5.3 Source quality hierarchy

The model's constants rest on a hierarchy of source quality:

1. **Primary population-PK** (TC): Bi et al. [1] supply CL/F, V/F, and K_a with RSEs and bootstrap CIs. This is the strongest source in the model.
2. **Primary single-dose PK** (TE): Ichihara et al. [2] supply terminal half-life and per-subject absorption rates in the target population (transgender men, IM route).
3. **Regulatory labels** (T gel, oral T, TU $t_{\max}$): AndroGel [7] and Jatenzo [6] supply steady-state C_{avg}, $C_{\max}$, and $t_{\max}$ from FDA registration trials. Aveed's FDA clinical pharmacology supplies TU's median $t_{\max}$.
4. **High-quality secondary compilation** (EV, EC): transfemscience.org [8] aggregates published single-dose and steady-state estradiol data. Not a primary table, and flagged as such.
5. **Secondary literature** (TU terminal half-life): Zhang et al. [3] and Wang et al. [4] supply terminal half-life and steady-state peak/trough, respectively. These two sources are in conflict (Section 5.1.1).
6. **Nothing** (TP; the spironolactone adjustment; the aromatization fraction): beyond the clearance identity that ties TP's magnitude to TC, there is no primary source for TP's absorption rate. The spironolactone clearance factor has no supporting human PK study and is mechanically misapplied (Section 2.16). The aromatization mass fraction $f_{\mathrm{arom}} = 0.002$ is a round number, conservative against the observed male E_2/T ratio. All three are flagged, all three are off or advisory by default, and none is presented to users as a fact about the drug.

5.4 The per-user calibration factor as a pragmatic IIV analog

The published population-PK models for testosterone esters all report substantial inter-individual variability. Ichihara et al. [2] report a CV of 0.58 on V_d/F for TE, and Bi et al. [1] report IIV terms on clearance and volume. A formal mixed-effects model would represent this variability with η terms on each parameter, drawn from a population distribution.

Isocline's calibration factor α is a simpler approach: a single scalar multiplier on the entire model output, derived from the user's own blood test. It cannot represent parameter-specific variability (e.g., a user with fast clearance but normal volume), and it cannot provide population-level confidence intervals. But it is practical for a consumer application: it requires no raw data access, no fitting infrastructure, and no statistical expertise from the user, and it absorbs the dominant source of individual-level error once a blood test is available.

The clamp range $[0.2, 5.0]$ spans most of the reported IIV distribution without admitting values that would indicate a data entry error. For routes with especially high between-subject variability—transdermal in particular—α is more important than on any other route.

6 Conclusion

We have presented the pharmacokinetic model underlying Isocline, a mobile application for hormone replacement therapy monitoring. The model employs a one-compartment Bateman biexponential with linear superposition of dose contributions, a dynamic endogenous baseline, and a per-user calibration factor. Ten medication/route/form combinations are supported across estradiol and testosterone.

The model's key structural features are:

- A **conservation-of-mass identity** that constrains the scaling constant S for all injectable testosterone esters to depend only on k_e and clearance, eliminating S as a free parameter and enforcing mutual consistency across esters.
- An **analytic closed-form derivation** of S for testosterone cypionate from primary population-PK parameters, which anchors the identity.
- A **per-user calibration factor** α that serves as a pragmatic analog of inter-individual variability terms in formal mixed-effects models.
- A **dynamic endogenous baseline** that models gonadal suppression over time, separating exogenous from endogenous contributions.

Every numerical claim in Tables 5, 7, 8, 9 and the persona suite was recomputed for this revision from the shipped constants, by forward simulation and—where a closed form exists—independently by the analytic steady-state sum. All but one reproduced; testosterone propionate's steady-state trough is corrected from 760 to 580 ng/dL. The aromatization equation is corrected by the factor of ten it had dropped in transcription.

The model achieves 0.0–0.1% accuracy against its anchors and 1–15% on out-of-sample checks. Seven of eight forms with a $t_{\max}$ are calibrated against primary or label sources; one (TP) is honestly flagged as uncalibrated. Structural limitations are quantified rather than hidden: TU's peak/trough ratio is a source conflict, not a structural deficiency; transdermal intraday ripple is under-modeled by construction; and oral peak fluctuation is under-predicted by an anchoring choice that prioritizes C_{avg}.

Three defects identified in this revision are of a different character, in that they admit repair rather than only disclosure. The endogenous baseline (Section 5.1.6) is keyed to the tracked hormone instead of the user's gonadal starting point, and is wrong for two of the three populations the application serves; the fix is a data-model field, not a pharmacokinetic one. The calibration factor (Section 2.15) does not round-trip, and Equation 10 states the derivation that would make it. The baseline taper's 92% plateau (Section 2.14) introduces a two-year discontinuity that removing one clamp would eliminate. Each is recorded here with its magnitude so that the correction can be checked against the claim.

The model is honest about where its constants are primary-sourced, where they are secondary, and where they remain uncalibrated. This honesty— the explicit marking of what is known, what is estimated, and what is placeholder—is, in our view, the most important property of a pharmacokinetic model used in a consumer health context, where users may make medical decisions based on its output.

This paper describes a consumer tracking application's internal model. It is not clinical guidance, and its predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.

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